TSA 2021

Problem Solving

Questions taken from the TSA Oxford 2021 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2021 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) The fairy lights on a Christmas tree are in three colours, blue, green and silver. The blue lights flash every 5 seconds, the green lights flash every 8 seconds and the silver lights flash every 12 seconds. The lights have all just flashed at the same time.

How many times will they all flash at the same time in the next 45 minutes?

A 5

B 22

C 45

D 108

E 112

2 (Q6) I have received the following message that is encrypted with a Caesar cipher:

JYMDWEFTQEBAF

This Caesar cipher replaces each letter of the plaintext (the message to be encrypted) with a letter that is a certain fixed number of positions along the 26-letter English alphabet. Knowing this fixed number of positions (the shift parameter) is the key to decrypt the message. For example, if the letter in the plaintext is A (the 1st letter of the alphabet) and the shift parameter is 4, then the encrypted letter is E (the 5th letter of the alphabet). Similarly, with a shift parameter of 7, Y (the 25th letter of the alphabet) is encrypted as F, which is the 32nd letter if the alphabet is repeated immediately after Z.

I would like to decrypt the message that I have received but I do not know the shift parameter. I only know that the first letter of the plaintext is X.

What is the sixth letter of the decrypted message?

A P

B Q

C R

D S

E T

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3 (Q7) Gordon's family are keen cyclists and regularly cycle around town (urban) and a nearby forest (rural). He has become concerned recently that the air they are breathing whilst cycling is causing his family respiratory problems. He has decided to buy his whole family anti-pollution cycling masks, by buying two adult and two child masks. Jo, his wife, has asthma which is caused by pollen and his son has allergies to clothing.

Information about five cycling masks is shown below in the table.

	Air-1	*GXi*	*Optiair*	*Speed*	*Breathe*
exhaust fume filtration	x	✓	✓	x	✓
replaceable filter system	✓	✓	x	✓	x
size: children	✓	x	✓	x	✓
size: adult	✓	✓	✓	✓	✓
dust and pollen filtration	✓	x	✓	✓	✓
hypoallergenic (kind to allergy sufferers)	✓	✓	x	✓	✓
urban environment	x	✓	✓	x	✓
rural environment	✓	x	✓	✓	✓
cold environment	✓	x	x	✓	x

Gordon wants to buy the brand of cycling mask which fits the whole family's requirements.

Which cycling mask should he buy?

A Air-1

B GXi

C Optiair

D Speed

E Breathe

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4 (Q8) A signal transmitter transmits a message in binary form. (In other words, the message is coded in 'bits'. Each bit is represented either by the digit '1' or by the digit '0'.)

To add an element of security, the signal is embedded within other random bits (so that there are other 1s and 0s before and after the message), creating a 26-bit transmission.

A particular 12-bit message is transmitted as follows:

01100010111001010011000100

The same message is transmitted again, as follows:

10101001111011100101000101

What is the 12-bit message contained within these two transmissions?

A 001010011000

B 001111011100

C 010011000100

D 101110010100

E 111101110010

5 (Q12) Help Yourself! is a restaurant in which customers can help themselves to as much food as they wish from a buffet for a fixed price of $10 for adults and $6 for children.

A Family Meal Ticket is available for $400 that allows unlimited visits to the restaurant for 2 adults and 2 children for one year. Any families with more than 2 children are charged half the children's price for every additional child, every time they eat.

Mr and Mrs Monroe and their 3 children made good use of their Family Meal Ticket last year, eating at Help Yourself! 30 times.

How much did the Family Meal Ticket save the Monroe family last year compared to if they had just paid per visit?

A $470

B $560

C $590

D $650

E $740

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6 (Q13) Jason's workplace offers a special deal on any four breakfast items. He can choose from bacon, sausage, eggs, toast, tomatoes and beans.

Jason will not have bacon and sausage together, nor will he have egg with beans.

How many different combinations of the six items will be acceptable to Jason?

A 2

B 3

C 4

D 5

E 6

7 (Q14) Cyclists in a small town were surveyed about their age. A table was produced to show the percentages and numbers of cyclists in five categories of their age. Some of the survey data was added to the table.

age categories	cumulative percentages of cyclists	numbers of cyclists
up to 10 years	15%	270
up to 18 years	50%	
up to 40 years	80%	
up to 60 years	90%	
up to 85 years	100%	

How many cyclists in the survey were above 40 years of age?

A 180

B 200

C 270

D 360

E 540

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8 (Q18) Tim and Jerry are business partners. Tim lives in New York (Eastern Time Zone) and Jerry lives in San Francisco (Pacific Time Zone). They both agree to meet in Oklahoma (Central Time Zone).

time zones relative to Pacific Time Zone		
Mountain	Central	Eastern
+1 hour	+2 hours	+3 hours

Tim set off at 10 am local time but shortly afterwards he realised he had left his ticket at home. As a result he missed his 4 hour flight to Oklahoma, so had to get the 12 pm (midday) flight instead.

Jerry travelled by train for 2 hours to Salt Lake City (Mountain Time Zone), leaving San Francisco at 6 am local time. He then immediately boarded a bus from Salt Lake City to Oklahoma, where he arrived 6 hours later.

Which one of the following is true?

A Jerry arrived in Oklahoma 3 hours before Tim.

B Jerry arrived in Oklahoma 1 hour before Tim.

C Jerry and Tim both arrived in Oklahoma at the same time.

D Tim arrived in Oklahoma 1 hour before Jerry.

E Tim arrived in Oklahoma 3 hours before Jerry.

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9 (Q19) Martha is hoping that her parents will agree to a family visit to Rockworld Zoo as a treat for her 16th birthday. This would include Martha, her mum and dad, her brothers aged 14, 12 and 9 years old and her grandmother, who is 62.

They have looked at the price tariff for tickets and have decided that they would want to book online the day before the proposed visit and that they could go either on Martha's birthday or on the day before it.

The table shows the possible ticket prices.

Type of ticket	*At the gate on the day*	*Booked online 1 day in advance*	*Booked online 7 days in advance*
Adult (16–59 years)	$17.00	$15.00	$14.00
Senior Citizen (60+)	$15.00	$13.00	$12.00
Child (3–15 years)	$13.00	$11.00	$10.00
Family of 4*	$54.00	$49.00	$46.00
Family of 5**	$63.00	$57.00	$54.00

* Family of 4 can be 2 adults and 2 children or 1 adult and 3 children
**Family of 5 can be 2 adults and 3 children or 1 adult and 4 children

What would be the smallest amount that the family could pay for their tickets?

A $76

B $81

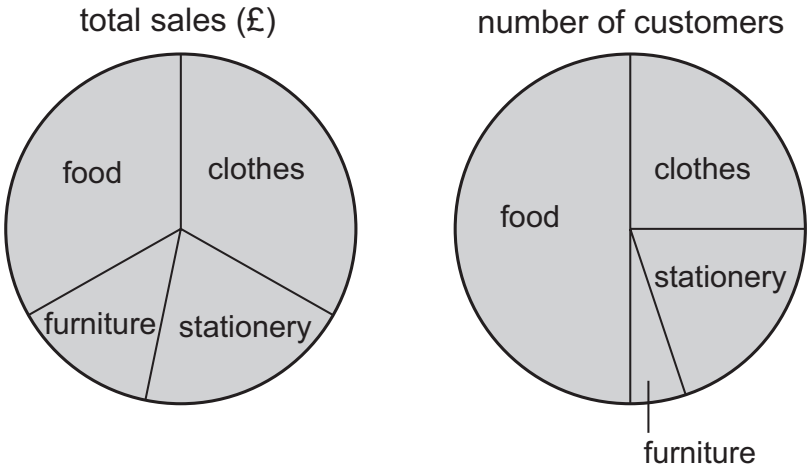
C $84

D $87

E $91

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10 (Q20) The sales for the four departments of a store are shown in the two pie charts below.



Which one of the following lists the departments in order, from highest average amount spent per customer to lowest?

A clothes, furniture, stationery, food

B food, clothes, stationery, furniture

C food, stationery, clothes, furniture

D food, stationery, furniture, clothes

E furniture, clothes, stationery, food

11 (Q24) A cinema uses magnetic letters to write the title of the most popular film that it is screening on its advertisement board. A set of magnetic letters includes the letters of the English alphabet in both upper and lower cases, i.e. 26 capital letters and 26 small letters. The cinema has just enough sets to be able to write the following title:

STAR WARS The Force Awakens

It is now considering putting this title up all in capital letters:

STAR WARS THE FORCE AWAKENS

What is the minimum number of extra sets of magnetic letters that the cinema needs to buy?

A 0

B 1

C 2

D 3

E 4

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12 (Q25) I run a bakery and sell two sizes of loaf. Small loaves (weighing 240 g) sell for 80p, and large loaves (weighing 400 g) sell for £1.20.

I buy dough, ready-made, at the start of each day. Small boxes (5 kg of dough) cost me £10.00 each, and large boxes (15 kg of dough) cost me £20.00 each.

The baking process causes each loaf to lose 20% of the weight of the unbaked dough.

I expect to sell 40 small loaves and 50 large loaves each day. I buy enough dough to be able to make these, and any unused dough at the end of the day has to be thrown away.

Assuming that I do not buy any more dough than necessary, what proportion of the dough that I buy each day do I have to throw away?

A 5.2%

B 6.3%

C 7.5%

D 17.8%

E 34.8%

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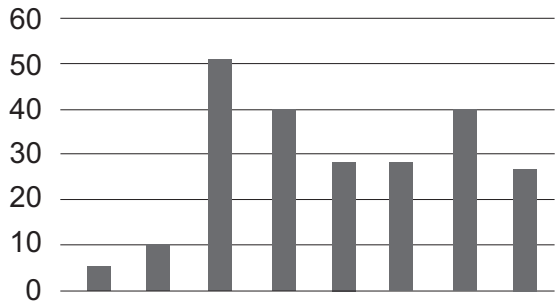
13 (Q26) The Zambezi River Basin plays a key role in water management for the following eight countries.

The table below shows the percentage of people with access to clean drinking water in each country in 2004.

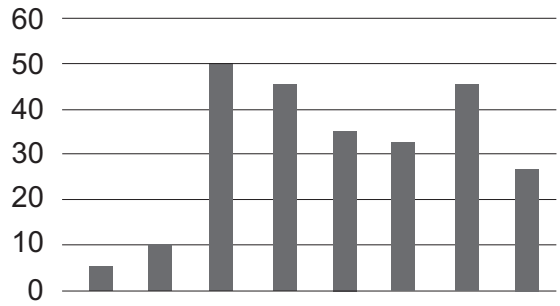
	percentage with access to clean drinking water (%)	
	urban	rural
Angola	40	34
Botswana	100	90
Malawi	95	44
Mozambique	81	41
Namibia	100	67
Tanzania	90	57
Zambia	88	48
Zimbabwe	100	73

Which one of the following graphs, suitably labelled, shows the difference in the percentages of people with access to clean water in urban versus rural areas in these countries?

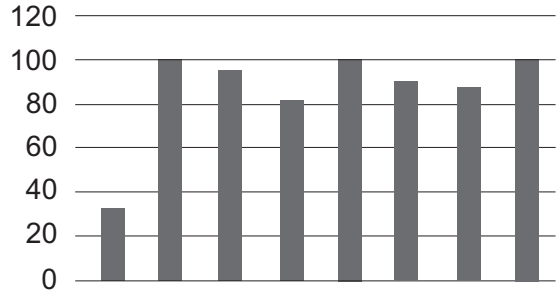
A



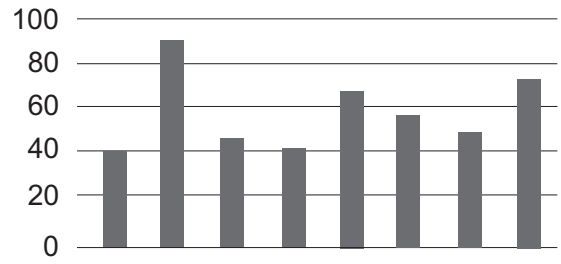
B



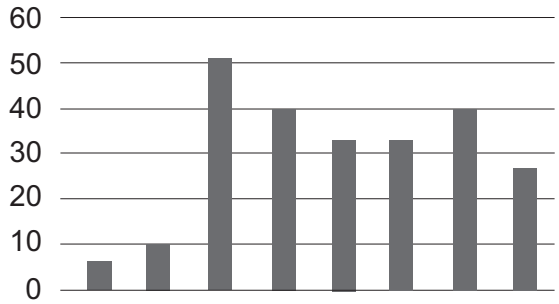
C



D



E



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14 (Q30) I am planning a party next weekend. I have invited 100 people. Last week I took advantage of an offer at my local supermarket where I was able to get 10% off the total price of party food if I spent more than $75. I paid $108 and I estimate that the food that I bought will be enough for 60 people. I've now had replies to the invitations and I know that there will be 80 people at the party. This afternoon I will buy the remaining food that I need.

What is the total price of the extra food that I need?

A $32.40

B $36.00

C $40.00

D $72.00

E $80.00

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15 (Q31) Aliyah is planning a 2 week holiday in Spain with her friend Kyra and her husband. They have chosen their hotel and are calculating what the total cost will be for the 3 of them. They have decided on a departure date of May 21st and will be ready to make the booking in early January.

The following table shows dates and prices.

All prices shown are £s per person.

duration	*Mar 29–Apr 22*	*Apr 23–29*	*Apr 30–May 6*	*May 7–13*	*May 14–June 10*	*June 11–17*	*June 18–24*
7 nights	£580	£600	£650	£670	£700	£700	£700
14 nights	£900	£900	£970	£1000	£1050	£1050	–
21 nights	£1200	£1230	£1300	£1350	£1400	–	–

Single room supplement: £11 on all dates.

Upgrade to a sea view room supplement: £3.

Supplement prices are per person, per night.

Early booking discount: if booked by 31st January is 5% up to a total of £2500, and 10% on any total above £2500.

Aliyah needs a single room. Kyra wants to upgrade the room for her and her husband so that it has a sea view.

What will be the total cost of the hotel booking for the 3 friends?

A £2181.20

B £2910.60

C £2973.60

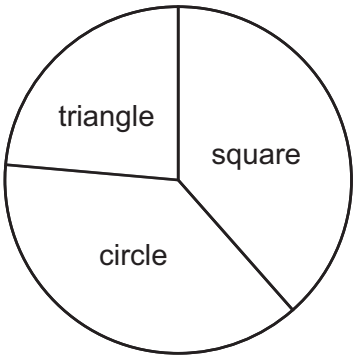
D £3049.20

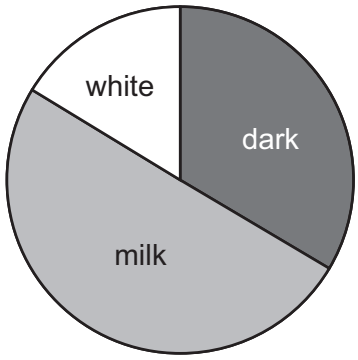
E £3218.60

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16 (Q32) A tin of 60 chocolates contains dark chocolates, milk chocolates and white chocolates. Each of these types of chocolate is present in three different shapes – square, circle and triangle.

The distributions of the chocolates in the tin, by type and shape, are shown in the pie charts:





Which one of the following tables, detailing the contents of the tin, is consistent with the pie charts?

A

	dark	milk	white
circle	8	10	2
square	8	14	4
triangle	4	6	4

B

	dark	milk	white
circle	5	10	5
square	8	10	2
triangle	8	6	6

C

	dark	milk	white
circle	5	11	6
square	5	11	6
triangle	5	8	3

D

	dark	milk	white
circle	8	8	4
square	6	12	2
triangle	5	10	5

E

	dark	milk	white
circle	9	10	4
square	9	10	4
triangle	2	10	2

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17 (Q36) A game is played involving two dice. Both dice are 6-sided cubes with the following shapes on their faces:

red square
blue square
green square
red circle
blue circle
green circle

The two dice are thrown and the player scores points according to the following system:

outcome	*points scored*
Each circle displayed	1
Each red shape displayed	1
Both same shape	1
Both same colour	2

The maximum possible score from a single throw of both dice is 7 (both dice displaying red circles).

Which one of the following scores is **not** possible from a single throw of both dice?

A 2

B 3

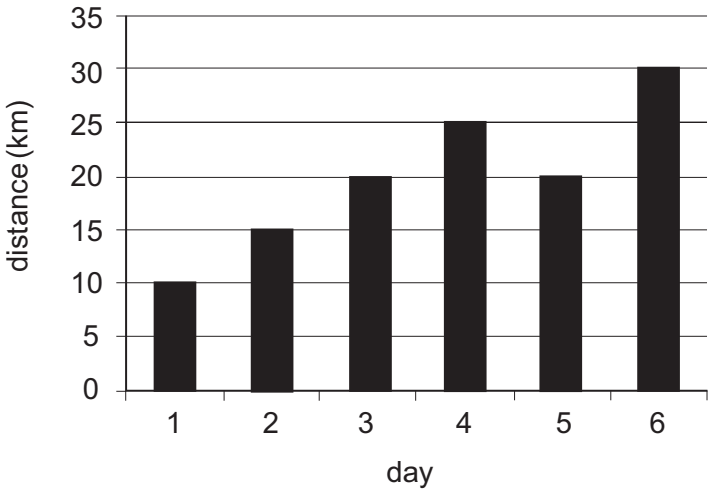
C 4

D 5

E 6

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18 (Q37) On a six-day hiking holiday, a group of friends travelled part of the trail by walking and part by bus. On average they walked 15 km per day over the six days. By the end of the 5th day, they had travelled 30 km by bus. The chart below shows the total distance covered each day.



How far did they travel by bus on the 6th day?

A 0 km

B 5 km

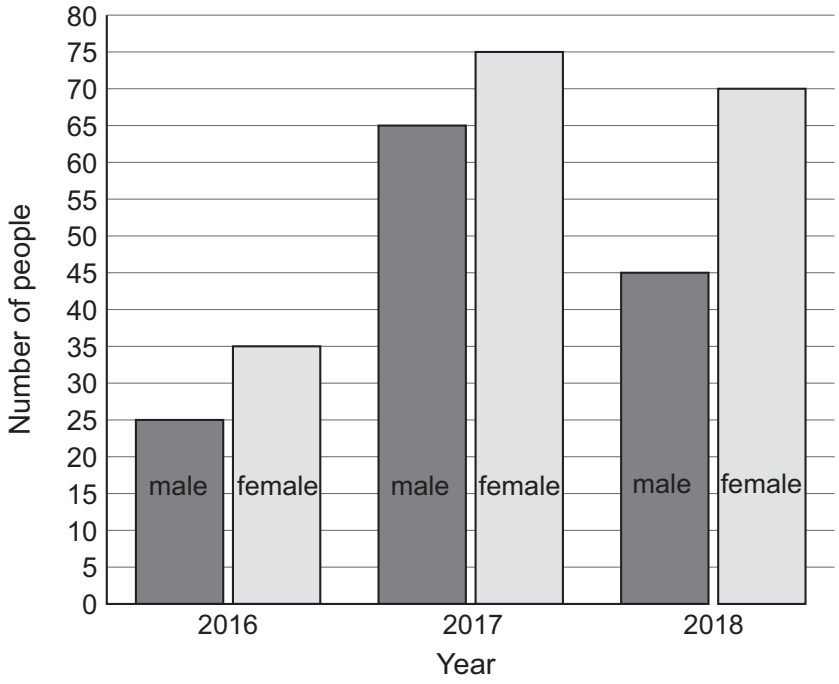
C 10 km

D 15 km

E 30 km

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19 (Q38) The numbers of males and females enrolled at a college in 2016, 2017 and 2018 are shown in the graph below.



Which one of the following is **not** true?

A The percentage of males increased from 2016 to 2017.

B The percentage of females increased from 2017 to 2018.

C The percentage of males increased from 2016 to 2018.

D The percentage increase in the number of males from 2016 to 2017 was greater than the percentage decrease in the number of males from 2017 to 2018.

E The percentage increase in the number of females from 2016 to 2017 was greater than the percentage decrease in the number of females from 2017 to 2018.

20 (Q42) The 4-digit PIN for my credit card can be written as two 2-digit square numbers (between 10 and 100). When the four digits, which are all different, are written individually in words, they are in alphabetical order.

What is the last digit of my PIN?

A 1

B 4

C 5

D 6

E 9

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21 (Q43) Jun lives in Applecross and Aki lives in Damsonville. Both live a 10-minute walk from their local train stations, which both lie on the Applecross–Figtown line.

Trains on this line run throughout the day to the following timetable:

Applecross–Figtown			
Applecross	h:07	h:27	h:47
Bananagate	h:17	h:37	h:57
Cherrytree	h:25	h:45	(h+1):05
Damsonville	h:34	h:54	(h+1):14
Elderhampton	h:48	(h+1):08	(h+1):28
Figtown	h:54	(h+1):14	(h+1):34

Figtown–Applecross			
Figtown	h:07	h:27	h:47
Elderhampton	h:13	h:33	h:53
Damsonville	h:27	h:47	(h+1):07
Cherrytree	h:36	h:56	(h+1):16
Bananagate	h:44	(h+1):04	(h+1):24
Applecross	h:54	(h+1):14	(h+1):34

“h” refers to all hours from 06:00 to 21:00. For instance, the train departing from Applecross at 08:27 arrives in Figtown at 09:14.

Jun leaves his house at 10:00 to go to visit Aki at her house. He must be back home by 16:00 and will spend at least 1 hour at Aki’s house.

What is the difference between the maximum and minimum time that Jun can spend at Aki’s house?

A 2 hours 40 minutes

B 2 hours 53 minutes

C 3 hours 26 minutes

D 3 hours 40 minutes

E 3 hours 53 minutes

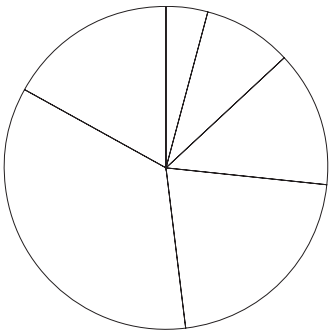
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22 (Q44) At the beginning of March, Suzy begins training for a marathon running event. At the end of each month from March until August, she records the total distance that she has run since the beginning of March. This information is shown in the following table.

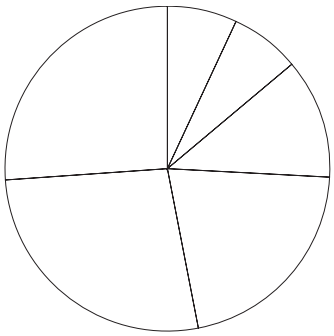
month	*total cumulative distance*
March	60 km
April	122 km
May	186 km
June	292 km
July	478 km
August	718 km

Which one of the following pie charts, suitably labelled, could represent a comparison of the monthly distances run by Suzy?

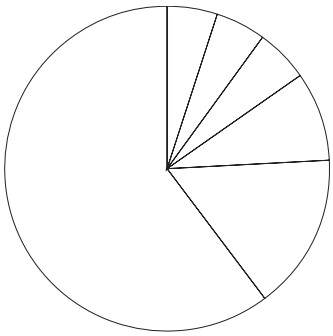
A



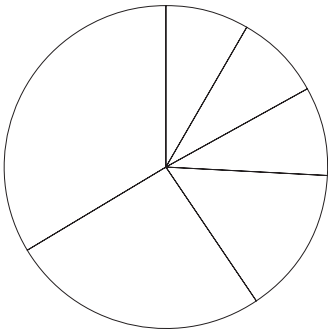
B



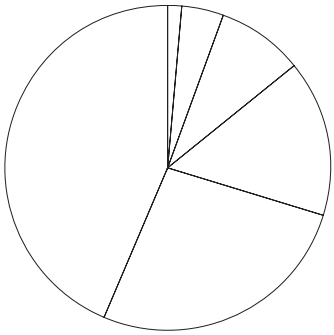
C



D



E



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23 (Q46) In August, Bill used his new, UK-based Bridgford Bank credit card for the first time. The summary details for this card are shown below.

Bridgford Bank Credit Card Summary Details

No interest for the first 12 months.

2.5% charged on cash withdrawals (minimum charge £3).

3.5% charged on all transactions made abroad (minimum £5).

All charges are added to the account at the time of the transaction and then the minimum monthly payment of 2% of the total shown on the statement is calculated at the end of each month.

He bought goods to the value of £2450. He also withdrew £300 in cash from a UK cash machine. Then, on holiday in Spain, he used the card to pay for £500 worth of goods.

What would be the minimum payment for his card use in August?

A £65.00

B £65.16

C £65.25

D £65.50

E £65.56

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24 (Q48) A company analyses five possible cities to build a 5 storey office building. The total floor area of the 5 storeys required by the company is 3000 m^2 and the total land area required on which to build it is 600 m^2.

The table below gives the land price and construction price in \$ per m^2 in five cities.

city	*construction costs* (\$ per m^2) *of total floor area*	*land purchase costs* (\$ per m^2)
Calypsio	1000	450
Jaspoon	2000	350
Menoko	1500	500
Sontorius	1300	400
Zentium	1600	350

The cost of building the office is the cost per square metre for the total floor space of all 5 storeys requirements and the plot land area purchase cost.

What is the difference in cost to build the office between the most expensive and least expensive of the five cities lowest and highest estimates for building the office?

A \$870 000

B \$1 210 000

C \$1 410 000

D \$2 700 000

E \$2 940 000

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25 (Q50) The company that I currently work for pays its employees monthly. Pay day is the fourth Thursday of every month or the 26th, whichever is earlier. Since I joined the company the dates on which I have been paid (in order) have been:

24th

24th

26th

26th

23rd

26th

25th (last month)

On what date will I be paid this month?

A 22nd

B 23rd

C 24th

D 25th

E 26th

END OF TEST

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